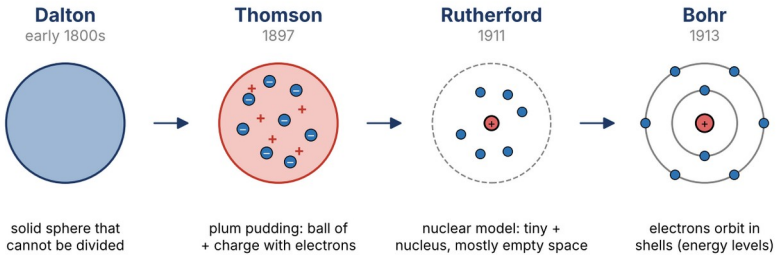
Development of the Model of the Atom

AQA GCSE Combined Science: Trilogy · Chemistry Topic 1 · Revision activity 3 · 19 marks · about 25 minutes

Read the key information (5 min), answer the **quick recall** questions without looking back (5 min), then answer the **exam questions** (15 min). Mark your work with the mark scheme on the last page.

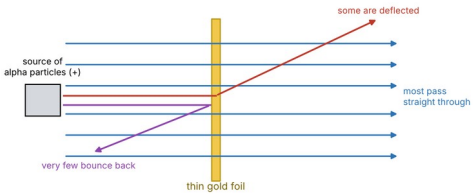
Key information

- Scientific models change when **new experimental evidence** is found that the old model cannot explain.
- **Dalton (early 1800s):** atoms were tiny solid spheres that could not be divided.
- **Thomson (1897):** discovered the **electron**, so atoms must contain smaller particles. He suggested the **plum pudding model** – a ball of positive charge with negative electrons embedded in it.
- **Rutherford (1909–1911):** the **alpha particle scattering experiment** (carried out by Geiger and Marsden) led to the **nuclear model** – a tiny, positive nucleus where most of the mass is, with electrons around it. The atom is mostly empty space.
- **Bohr (1913):** electrons orbit the nucleus at **fixed distances** (shells / energy levels). His calculations agreed with experiments.
- Later experiments showed the positive charge of the nucleus is made of smaller particles called **protons**.
- **Chadwick (1932):** showed that the nucleus also contains **neutrons**.



The alpha particle scattering experiment

Positive alpha particles were fired at a very thin sheet of gold foil.



What the results showed

Observation	What it tells us
Most alpha particles went straight through	The atom is mostly empty space
Some alpha particles were deflected (changed direction)	The centre of the atom (nucleus) has a positive charge that repels them
Very few alpha particles bounced back	Most of the mass is concentrated in a very small nucleus

Section A: Quick recall (8 marks – 1 mark each)

1 Which scientist described atoms as tiny spheres that could not be divided?

2 Which particle did Thomson discover?

3 What is the name of the model of the atom suggested after the electron was discovered?

4 In the scattering experiment, what were alpha particles fired at?

5 Which scientist suggested that electrons orbit the nucleus in shells?

6 Which particle did James Chadwick discover?

7 Why do scientific models change over time?

8 Put these scientists in time order, earliest first: Bohr, Dalton, Chadwick, Thomson, Rutherford.

Section B: Exam questions

01 Describe the plum pudding model of the atom.

[2 marks]

02 In the alpha particle scattering experiment, most alpha particles passed straight through the gold foil, but a very small number bounced back.

02.1 What does it tell you that most alpha particles passed straight through?

[1 mark]

02.2 Explain why a very small number of alpha particles bounced back.

[2 marks]

03 Give two differences between the plum pudding model and the nuclear model of the atom.

[2 marks]

04 Explain why the plum pudding model was replaced by the nuclear model.

[2 marks]

05 Bohr's model of the atom was accepted by other scientists. Give two pieces of information about the atom that were discovered after Bohr's model.

[2 marks]

Mark Scheme – Development of the Model of the Atom

(1) = one mark. / separates acceptable alternatives. Words in brackets are not needed for the mark.

Question	Marking guidance	Comments	Mark
A	**1** Dalton **2** the electron **3** the plum pudding model **4** thin gold foil **5** Bohr **6** the neutron **7** new experimental evidence is found **8** Dalton, Thomson, Rutherford, Bohr, Chadwick	1 mark each	8
01	a ball / sphere of positive charge (1) with (negative) electrons embedded / scattered in it (1)		2
02.1	the atom is mostly empty space (1)		1
02.2	the mass is concentrated in a small nucleus (1) the nucleus is positive so it repels the positive alpha particles (1)		2
03	any two from: (1) each • nuclear model has a nucleus; plum pudding has no nucleus • in the nuclear model the positive charge is in a small nucleus; in plum pudding it is spread out • in the nuclear model the mass is concentrated in the nucleus; in plum pudding it is spread out • nuclear model is mostly empty space; plum pudding is not		2
04	results of the alpha scattering experiment could not be explained by the plum pudding model (1) e.g. alpha particles bouncing back needed a small, concentrated, positive nucleus (1)		2
05	the positive charge of the nucleus is made of protons (1) the nucleus also contains neutrons (discovered by Chadwick) (1)		2

Total: 19 marks